

Problem Set #2: HANK-SAM

PSE APE M2 – Quantitative Macroeconomics II

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Question 1: Solving and calibrating the steady state

1.a) Parameters

We set the parameters as established in the Problem Set statement.

1.b) Household Problem

From the Household's budget constraint we can derive an initial guess for b' (`bp_old`):

$$c = y(x) + rb - b', \quad c = y(x) \implies b' = rb \quad (1)$$

Our Endogenous Grid Method (EGM) loop converges in 270 iterations. We present the resulting plots for the Policy Functions in Figure 1. As we see, employed individuals (panel **a**, in blue) consistently decide on *accumulating* assets, with b' being above the 45° line ($b' > b$) for most of the asset grid, to insure themselves against the future possibility of unemployment. We find that the upper bound of $b_{max} = 5$ is binding in steady state, as $b = 5 \implies b' = 5$. In order to satisfy the Problem Set's requirement to choose a b_{max} 'large enough that the upper bound does not bind in equilibrium', we must extend our grid. The 'target wealth' for an infinitely-employed worker lies beyond b_{max} . We find the distance from b is relatively small, which makes sense given that the probability of becoming unemployed is quite low.

On the contrary, unemployed individuals (panel **a**), in red), *disaccumulates* (i.e., consumes) their current stock of assets ($b' < b$). The income of the unemployed ($z = 0.3$) is quite lower than the employed's wage ($w = 0.98$). The unemployed face a probability of $f = (\frac{\sigma}{u_{ss}}) - \sigma = 0.27$, and decide to smooth their consumption to prevent it from drastically falling to z , expecting to become employed at some point.

These dynamics are confirmed by the Consumption Policy Functions (panel **b**). An unemployed individual with the lowest, binding stock of assets ($b_{min} = -3$), cannot consume more than their current income $z = 0.3$, and can progressively consume more when they have a larger

stock of b . The employed individual can consistently consume more than the unemployed one, and even when constrained by b_{min} , they consume less than their current income $w = 0.98$ as to accumulate assets to insure from future income shocks.

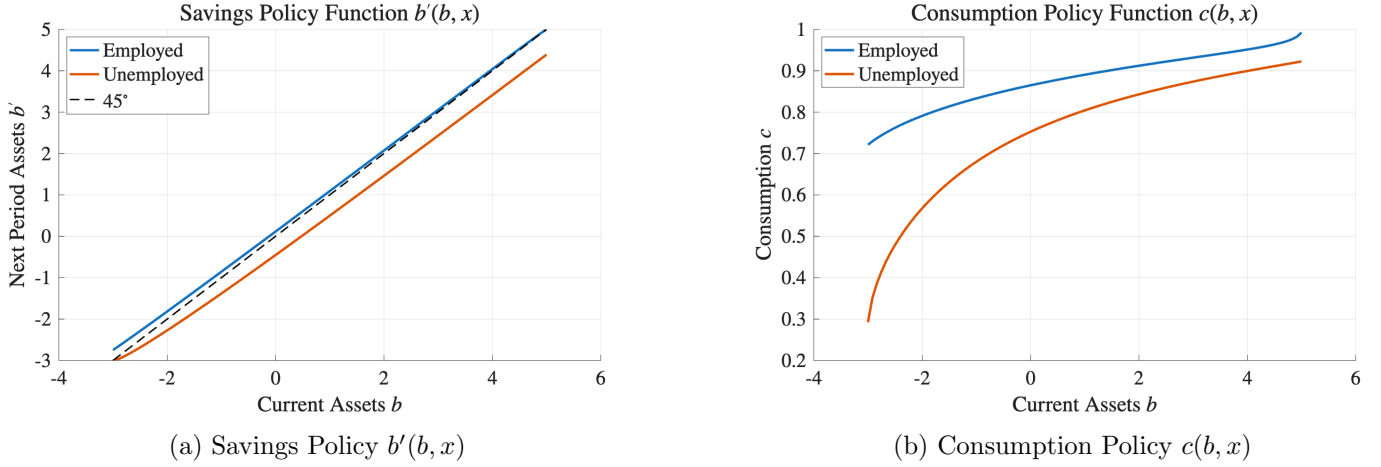


Figure 1: Steady State Policy Functions.

The binding and the non-binding equilibrium b_{max}

To verify this requirement and ensure the precautionary savings motive can fully play out, we solve the problem again by setting $b_{max} = 10$. This b_{max} setting permits the upper constraint being indeed non-binding. This will become more evident in the next section of our answers.

1.c) Stationary Distribution

To find the stationary joint distribution of assets and employment status, $g(b, x)$, we use the non-stochastic simulation method of Young (2010). We compute this distribution for both our baseline grid ($b_{max} = 5$) and our extended, non-binding grid ($b_{max} = 10$).

Unemployment Rate Verification

The steady-state unemployment rate implied by our discrete stationary distribution is calculated by summing the mass of all agents in the unemployed state: $u = \sum_{i=1}^N g_{i,u}$. For both grid specifications, our code yields an implied unemployment rate of exactly **0.1** (10%). This perfectly matches the theoretical formula provided.

Distribution Statistics

Table 1 summarizes the mean assets and the share of households constrained at the borrowing limit ($b_{min} = -3$) for both scenarios.

Statistic	Baseline ($b_{max} = 5$)	Extended ($b_{max} = 10$)	Final ($b_{max} = 24$)
Mean Assets (Overall)	2.8224	3.3675	3.5070
Mean Assets (Employed)	2.9514	3.5001	3.6397
Mean Assets (Unemployed)	1.6607	2.1741	2.3124
Share at Borrowing Constraint (Employed)	0.04%	0.04%	0.04%
Share at Borrowing Constraint (Unemployed)	0.97%	0.89%	1.02%

Table 1: Stationary Distribution Statistics across different upper bound choices.

As expected, employed workers hold significantly higher average assets than unemployed workers, reflecting their active accumulation of a *buffer stock*. When workers lose their jobs, they progressively spend their saved assets. Given the latter, the share of unemployed individuals hitting the strict lower borrowing constraint ($b_{min} = -3$) is substantially higher (almost 1%) than that of the employed (0.04%). These constrained unemployed workers are effectively “hand-to-mouth”, forced to consume only their unemployment income $z = 0.3$.

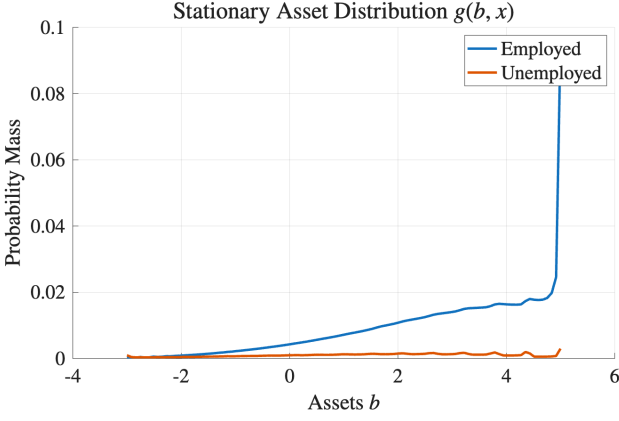
The Impact of the Upper Bound (b_{max})

The visual representation of these distributions in Figure 2 clearly illustrate why we chose to verify the results of $b_{max} = 5$.

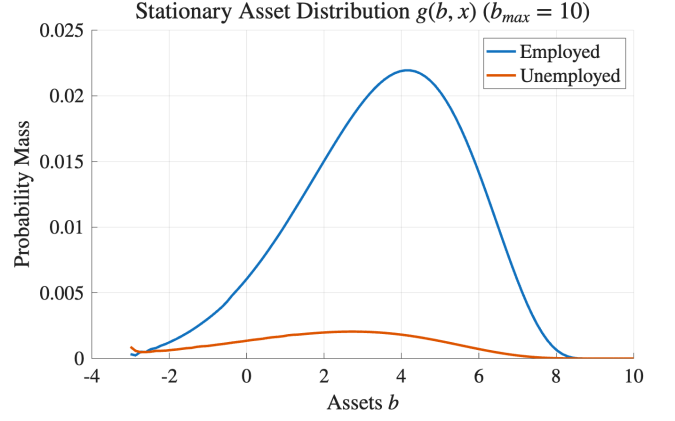
In panel (a), we observe bunching at the far right of the grid. Because the employed workers’ target *optimal* wealth lies beyond $b = 5$, their savings policy is clamped by the grid, forcing a large mass of the population (nearly 9.6%) to artificially accumulate at the boundary.

By extending the grid to $b_{max} = 10$ in panel (b), we allow the precautionary savings motive to fully play out. The distribution now smoothly “dies off” before hitting the upper bound, confirming that $b_{max} = 10$ represents a truly non-binding equilibrium. We further confirm this by verifying that $\max(b'_{policy}|x = e) < b_{max} = 10$. Without the cap at $b = 5$, the employed households are able to accumulate their desired stock, which drives the overall mean assets of the economy significantly higher (from 2.82 to 3.36).

Important note: In the rest of the Problem Set, we will proceed by using $b_{max} = 24$, as we received an e-mail stating so.



(a) Distribution with $b_{max} = 5$ (Binding)



(b) Distribution with $b_{max} = 10$ (Non-Binding)

Figure 2: Stationary Asset Distributions $g(b, x)$ for both upper bound specifications.

1.d) Labour market objects and calibration of κ

First, we can solve for m given the vacancy filling rate:

$$f(\theta) = m\theta^{1-\alpha} \implies m = \left(\frac{1}{\theta^{1-\alpha}}\right)\left(\frac{\sigma}{u_{ss}} - \sigma\right) \quad (2)$$

Then, because all parameters on the RHS are given, we know $m = 0.2700$. We then proceed as follows:

$$q(\theta) = m\theta^{-\alpha} \implies q(\theta) = 0.2700 \quad (3)$$

$$J = \frac{A - w}{1 - \beta_f(1 - \sigma)} \implies J = 0.5859 \quad (4)$$

$$\kappa = \beta_f q(\theta) J \implies \kappa = 0.1575 \quad (5)$$

We are effectively performing a “reverse engineering” calibration. By targeting $u_{ss} = 10\%$, $\sigma = 0.03$ and $\theta = 1$, we implicitly determine the job-finding rate f_β . We observe that the Profit Margin ($A - w$) acts as a target. Consequentially, this lets us analytically *calibrate* for the parameters dictating that our *targets* are met: the efficiency of the Matching Function (m), and the vacancy cost for the Free Entry Condition (κ).

In other words, m lets us pin the job-finding rate f such that the market tightness θ (the number of vacancies posted for every unemployed individual) is exactly one given the transition probabilities Π and the desired u_{ss} . Meanwhile, κ ensures that the cost of a vacancy equals the expected benefit of a filled job, as in (5).

1.e) Closing the asset market

The final step in our steady-state calibration is to ensure that the asset market clears. Theoretically, the aggregate demand for bonds, B^d , is the integral of all individual asset holdings across the state space. Given our discrete grid of $N_b = 100$ asset levels and two employment states (e, u), we can express this the following double sum:

$$B^d = \sum_{i=1}^N \sum_{j=1}^2 b_i g_{i,j} \quad (6)$$

In our numerical implementation (`main1_steady.m`), we compute this using the stationary distribution obtained from our non-binding baseline update of $b_{max} = 24$:

```
% MATLAB implementation of the aggregate bond demand
Bd = sum(bgrid_3 .* g_ss_3, 'all');
Bs = Bd;
```

Here, $\mathbf{b}$ is a column vector of asset levels and $\mathbf{g}$ is the $N \times 2$ matrix of probability masses. MATLAB is performing element-wise multiplication of the grid across both employment-state columns, and sums the entire result. Based on our results, the aggregate bond demand is **3.5070**.

Equilibrium Condition and Consistency of r

We let B^s denote the fixed supply of bonds in the model and impose the market-clearing condition:

$$B^s = B^d = 3.5070 \quad (7)$$

Choosing B^s this way ensures that the targeted steady-state real interest rate r is consistent with equilibrium by design. In a standard general equilibrium model, r is an endogenous price that adjusts until bond demand equals a fixed bond supply. However, since we have targeted a specific value for it ($r = 1 + 0.03/12$), we have “reverse-engineered” the supply side. By setting B^s exactly equal to the demand generated at our targeted r , we guarantee that the bond market clears at that specific interest rate without the need for further price adjustments.

The steady state is solved: the labor market is in equilibrium via the calibrated m and κ , households are optimizing their savings via EGM policy functions, and the asset market clears at our targeted interest rate.

Question 2: Recession experiment: permanent financial shock

2.a) General Equilibrium Loop and New Steady State

To find the new equilibrium in the context of a permanent financial shock ($b_{min} = -3 \rightarrow b'_{min} = -2.5$), we implement an iterative algorithm that updates prices and household policies until the bond market clears. The process follows these sequential steps:

1. **Initialize:** Guess an initial real interest rate, r , starting from our calibrated steady state value r_{css} .

2. **Update Monetary Variables and Wages:** Given r , we solve for the inflation rate π by combining the Fisher equation and the Taylor rule:

$$\begin{aligned} i = r\pi, \quad i = i_{ss} \left(\frac{\pi}{\pi_{ss}} \right)^{\phi_\pi} \epsilon &\implies r\pi = r_{ss}\pi_{ss} \left(\frac{\pi}{\pi_{ss}} \right)^{\phi_\pi} \epsilon \\ \implies \frac{r}{r_{ss}\epsilon} = \left(\frac{\pi}{\pi_{ss}} \right)^{\phi_\pi - 1} &\implies \pi = \pi_{ss} \left(\frac{r}{r_{ss}\epsilon} \right)^{\frac{1}{\phi_\pi - 1}} \end{aligned} \quad (8)$$

With π determined, the nominal interest rate i and real wage w naturally follow:

$$i = r\pi \quad (9)$$

$$w = w_{ss} \left(\frac{\pi}{\pi_{ss}} \right)^{-\xi} \quad (10)$$

3. **Update Labor Market Equilibrium:** Given the newly updated wage w , we recompute the labor market equilibrium values. First, we compute the firm's job value J :

$$J = \frac{A - w}{1 - \beta_f(1 - \sigma)} \quad (11)$$

Using the free entry condition, we solve for the labor market tightness θ :

$$\kappa = \beta_f q(\theta) J \implies \theta = \left(\frac{\kappa}{\beta_f m J} \right)^{-\frac{1}{\alpha}} \quad (12)$$

Then, we compute the vacancy filling rate $q(\theta) = m\theta^{-\alpha}$ and the job finding rate f :

$$f = m\theta^{1-\alpha} \quad (13)$$

These values determine the updated transition matrix Π for employment risk in the household problem.

4. **Solve Household Problem:** Given the new prices (r, w) and transition probabilities (f) , we solve for the new optimal saving policy $b' = h(b, x)$ on the updated grid $b'_{min} \leq b \leq b_{max}$ using EGM, as in Question 1.
5. **Compute Stationary Distribution:** We use the new policy function to simulate the stationary distribution of households $g(b, x)$ via the Young (2010) non-stochastic method.
6. **Check Market Clearing and Update r :** We compute aggregate bond demand $B^d = \sum_{i=1}^N \sum_{j=1}^2 b_i g_{i,j}$ and evaluate the error $err = B^s - B^d$. If $|err|$ is outside our tolerance, we update the interest rate via $r_{new} = r + \lambda \cdot err$ and repeat steps 2 through 6 until convergence.

Running this algorithm, the General Equilibrium (GE) loop converges to the new steady state in exactly **27 iterations**. The resulting macroeconomic variables are summarized in Table 2.

Variable	Original Steady State	New Steady State
Real Interest Rate (r)	1.002500	1.001578
Inflation (π)	1.001667	0.999824
Nominal Interest Rate (i)	1.004171	1.001402
Real Wage (w)	0.980000	0.982168
Market Tightness ($\theta = v/u$)	1.000000	0.794995
Job Finding Rate (f)	0.270000	0.240739
Unemployment Rate (u)	10.00%	11.08%

Table 2: Macroeconomic variables before and after the financial shock.

We note that $b_{max} = 24$.

2.b) Distributional Changes

The financial shock generates significant changes in the distribution of household assets. Table 3 compares the asset distribution statistics between the original steady state and the new equilibrium.

Statistic	Original Steady State ($b_{min} = -3$)	New Steady State ($b'_{min} = -2.5$)
Mean Assets (Overall)	3.5070	3.5070
Mean Assets (Employed)	3.6397	3.6677
Mean Assets (Unemployed)	2.3124	2.2177
Share at Borrowing Constraint (Employed)	0.04%	0.06%
Share at Borrowing Constraint (Unemployed)	1.02%	1.62%

Table 3: Changes in average assets and the share of constrained households.

- **Average Assets:** In the original steady state, the aggregate bond demand was 3.5070. In the new steady state, despite the tighter constraint, the equilibrium real interest rate adjusts so that aggregate bond demand continues to equal the fixed bond supply ($B^s = 3.5070$) to clear the market. However, the composition of these assets shifts: employed workers accumulate slightly more precautionary savings (from 3.6397 to 3.6677), while the unemployed hold fewer assets on average (falling from 2.3124 to 2.2177).
- **Borrowing Constraint:** The share of unemployed households hitting the borrowing constraint significantly increased from 1.02% to 1.62%. Even the employed share at the constraint increased slightly (from 0.04% to 0.06%).

Visually, the left tail of the asset distribution $g(b, x)$ becomes compressed at the new $b'_{min} = -2.5$, pushing a larger mass of the unemployed population directly against the new, stricter limit. Figure 3 overlays the distributions before and after the shock, clearly illustrating the inward shift of the asset holdings and the change in conditional means.

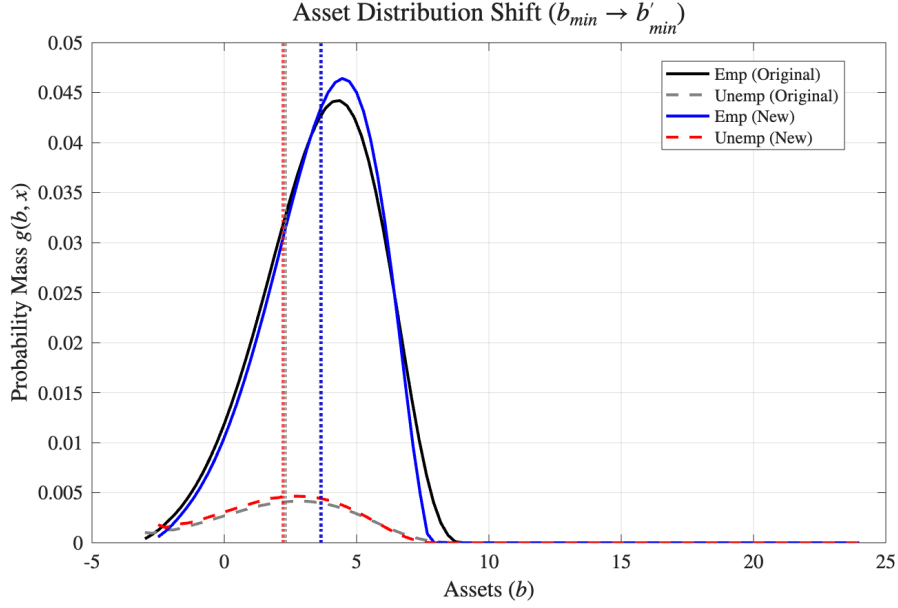


Figure 3: Stationary Asset Distributions $g(b, x)$ before and after the financial shock. Vertical dotted lines represent the conditional mean assets for each state. *Note:* $b_{min} = -3$ while $b'_{min} = -2.5$.

2.c) Economic Mechanism and Amplification

The transition from the financial shock to the rise in unemployment operates through a combination of precautionary savings, aggregate demand effects, and nominal rigidities.

1. Precautionary Savings and the Real Rate: When the borrowing constraint tightens, households face higher uninsurable idiosyncratic risk. To maintain a buffer away from the new limit, their precautionary demand for savings increases. Because the supply of bonds B^s is fixed, this excess demand for assets forces the real interest rate r to fall (from 1.0025 to 1.0015) to clear the bond market.

2. Aggregate Demand and Inflation: The nominal block connects the real rate to inflation via the Fisher equation ($i = r\pi$) and the Taylor rule. Because the central bank reacts strongly to inflation ($\phi_\pi = 1.5$), a lower real equilibrium interest rate requires a lower steady-state inflation rate ($\pi = 0.9998$, meaning deflation) to satisfy both rules simultaneously. This represents a drop in aggregate demand.

3. Wage Rigidity and the Labor Market: If nominal wages were perfectly flexible, the drop in inflation would not affect the real economy. However, because nominal wages are sticky ($\xi = 1.2$), the fall in prices mechanically drives up the *real* wage w from 0.9800 to 0.9821.

4. Firm Free Entry: The higher real wage compresses the firm's surplus ($A - w$). Consequently, the value of a filled job (J) drops. Under the free-entry condition, firms respond by posting fewer vacancies, causing market tightness ($\theta = v/u$) to fall from 1.0 to 0.795. This makes it significantly harder for workers to find jobs (f falls to 0.24), finally driving the unemployment rate up to 11.08%.

To understand the magnitude of this response, we alter the parameters governing the Taylor

rule (ϕ_π) and nominal wage rigidity (ξ).

2.d) The Role of Parameters: ϕ_π and ξ

To thoroughly understand how nominal rigidities and monetary policy amplify the financial shock, we repeat the general equilibrium procedure under four alternative parameter specifications. The resulting macroeconomic variables are summarized in Table 4.

Variable	Original SS	Flexible Wages ($\xi = 0$)	High Rigidity ($\xi = 1.5$)	Aggressive CB ($\phi_\pi = 2.0$)	Passive CB ($\phi_\pi = 1.2$)
Real Interest Rate (r)	1.002500	1.002156	0.996063	1.002019	0.999221
Inflation (π)	1.001667	1.000979	0.988845	1.001186	0.985391
Nominal Interest Rate (i)	1.004171	1.003137	0.984953	1.003207	0.984623
Real Wage (w)	0.980000	0.980000	0.999122	0.980565	0.999456
Market Tightness (θ)	1.000000	1.000000	0.001929	0.944340	0.000740
Job Finding Rate (f)	0.270000	0.270000	0.011858	0.262378	0.007347
Unemployment Rate (u)	10.00%	10.00%	71.67%	10.26%	80.33%

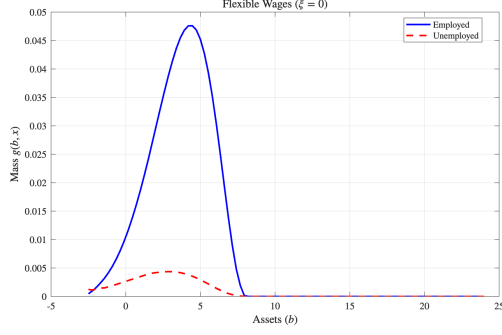
Table 4: Macroeconomic outcomes under varying degrees of nominal rigidity and monetary policy responsiveness.

The Role of Wage Rigidity (ξ): When nominal wages are perfectly flexible ($\xi = 0$), the classical dichotomy holds. Despite the drop in aggregate demand and subsequent deflation (π drops to 1.0010), the real wage remains perfectly anchored at 0.9800. Consequently, firm surplus does not shrink, vacancy posting remains stable, and the unemployment rate is perfectly insulated, remaining at 10.00%. Conversely, under high wage rigidity ($\xi = 1.5$), the deflationary pressure forces the real wage to spike to 0.9991, nearly eliminating firm surplus ($A = 1$). This reduces market tightness and causes unemployment to rise significantly to 71.67%.

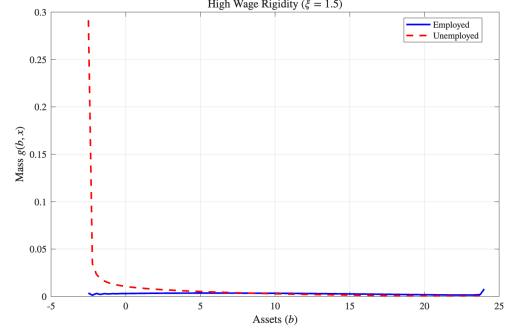
The Role of the Taylor Rule (ϕ_π): The central bank's responsiveness dictates how much deflation is required to achieve the new, lower market-clearing real interest rate. An aggressive central bank ($\phi_\pi = 2.0$) cuts the nominal interest rate rapidly in response to falling prices. This requires only a small drop in inflation to achieve the necessary real rate, mitigating the upward pressure on real wages and keeping unemployment relatively stable at 10.26%. A passive central bank ($\phi_\pi = 1.2$), however, cuts nominal rates too slowly. To force the real rate down, the economy must undergo massive deflation ($\pi = 0.9854$). Combined with sticky wages, this severe deflation drives the real wage up to 0.9995, resulting in a severe recession with 80.33% unemployment.

Distributional Impacts of Amplification

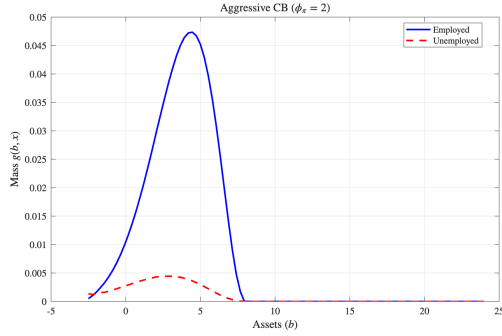
The severity of the macroeconomic collapse in the highly rigid and passive scenarios drastically alters household savings behavior. Figure 4 displays the final stationary distributions for all four cases.



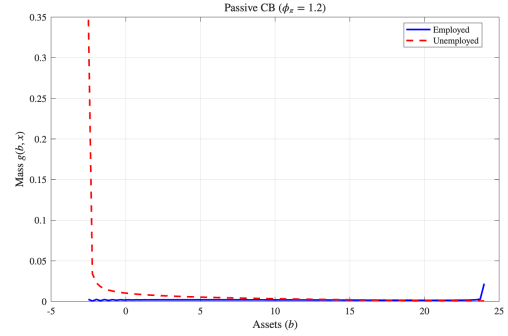
(a) Flexible Wages ($\xi = 0$)



(b) High Wage Rigidity ($\xi = 1.5$)



(c) Aggressive CB ($\phi_\pi = 2.0$)



(d) Passive CB ($\phi_\pi = 1.2$)

Figure 4: Stationary Asset Distributions $g(b, x)$ across the four sensitivity scenarios.

In the stable scenarios (panels a and c), the distributions closely resemble our baseline shock. However, in the extreme recession scenarios (panels b and d), the large spike in unemployment drastically shifts the population mass. Because up to 80% of the economy is unemployed and consuming their buffer stocks, a massive concentration of households becomes stacked firmly against the borrowing constraint.

Question 3: Monetary and fiscal policy responses to the shock

3.a) Offsetting the Financial Shock with Monetary Policy

Deriving the Optimal Monetary Policy Shock (ϵ)

To completely insulate the real economy from the permanent financial shock (the tightening of the borrowing constraint to $b'_{min} = -2.5$), the central bank must stabilize the labor market.

Specifically, to prevent a recession, the labor market tightness θ must remain at its steady-state value of 1, which implies an unchanged unemployment rate ($u = 10\%$) and an unchanged firm surplus ($A - w$). This requires the real wage w to remain at its original steady-state level (w_{ss}).

From the wage rigidity mechanism, we know that:

$$w = w_{ss} \left(\frac{\pi_{ss}}{\pi} \right)^\xi \quad (14)$$

For w to equal w_{ss} , inflation must be completely stabilized at its steady-state target: $\pi = \pi_{ss}$.

However, because the tighter borrowing limit significantly increases households' aggregate demand for precautionary savings, the asset market (where bond supply B^s is fixed) cannot clear at the original steady-state real interest rate (r_{ss}). To properly equate savings demand to supply under the new constraint, the “natural” real interest rate of the economy must fall to some $r^* < r_{ss}$, thereby decreasing the demand for bonds.

The central bank controls the nominal interest rate i via a Taylor rule augmented with a monetary policy shock ϵ :

$$i = i_{ss} \left(\frac{\pi}{\pi_{ss}} \right)^{\phi_\pi} \epsilon \quad (15)$$

Under optimal stabilization ($\pi = \pi_{ss}$), the Fisher equation gives $i = r^* \pi_{ss}$. Substituting this into the Taylor rule evaluations at steady-state values ($i_{ss} = r_{ss} \pi_{ss}$) yields:

$$r^* \pi_{ss} = r_{ss} \pi_{ss} \left(\frac{\pi_{ss}}{\pi_{ss}} \right)^{\phi_\pi} \epsilon \implies r^* \pi_{ss} = r_{ss} \pi_{ss} \epsilon \implies \epsilon = \frac{r^*}{r_{ss}} \quad (16)$$

Thus, to prevent the recessionary loop entirely, the central bank must calculate the new natural rate r^* and execute an accommodative monetary shock $\epsilon < 1$ that perfectly aligns the nominal trajectory with the required real interest rate.

Computational Strategy and Pseudo-Code

To computationally find the necessary ϵ and verify its effectiveness, we design an algorithm (`main3.epsilon.m`) that operates in three distinct phases:

1. Find the Natural Rate (r^*)

- We artificially “freeze” the labor market and real wages at their initial steady-state values ($u = 10\%$, $w = w_{ss}$, $\theta = 1$). We fix $\pi = \pi_{ss}$.
- We set up the household EGM problem with the newly tightened borrowing limit ($b'_{min} = -2.5$).
- We loop over guesses for the real interest rate, solving for the optimal policy functions and simulating the stationary distribution $g(b, x)$ until the aggregate bond demand B^d exactly matches B^s .
- The r that converges this loop is stored as r^* .

2. Extract the Taylor Shock (ϵ)

- Compute the optimal monetary policy shock analytically using our derived formula:
 $\epsilon = r^* / r_{ss}$.

3. Verify the General Equilibrium

- We initialize the full General Equilibrium loop (the same one used in Question 2), unfixing the labor market so that w , θ , and f are endogenous again.
- We embed the calculated ϵ directly into the Taylor rule.

- By solving the full model, we confirm that the mechanism converges perfectly to fully stabilized outcomes.

Results and Discussion

Phase 1 converges. Under the stricter constraint of -2.5 , precautionary savings behaviors increase bond demand. To clear the market, the required natural real interest rate r^* falls from 1.002500 to 1.002156 (10 iterations).

The optimal Taylor rule shock needed to enact this shift is calculated as $\epsilon = \frac{1.002156}{1.002500} = 0.999657$.

By embedding this exact ϵ into the full General Equilibrium model, Table 5 demonstrates the outcome of the stabilization policy compared to the baseline financial shock recession:

Variable	Unchecked Shock (Q2)	Accommodated Shock ($\epsilon = 0.999657$)
Real Interest Rate (r)	1.001578	1.002156
Inflation (π)	0.999824	1.001667
Nominal Interest Rate (i)	1.001402	1.003826
Real Wage (w)	0.982168	0.980000
Market Tightness ($\theta = v/u$)	0.794995	1.000000
Job Finding Rate (f)	0.240739	0.270000
Unemployment Rate (u)	11.08%	10.00%

Table 5: Macroeconomic outcomes underlying the central bank’s optimal accommodation policy compared to the unchecked recession mechanism.

Note: Post-policy distributions were found in 13 GE iterations.

Discussion: The results indicate that accommodative monetary policy completely insulates the real economy from the financial shock. Because ϵ mirrors the exact drop in the natural real interest rate, the central bank immediately avoids the necessity for deflation to clear the bond market. Because inflation stays perfectly anchored at π_{ss} , the real wage w never surges. Firms maintain their full profit margin ($A - w$), posting just enough vacancies ($\theta = 1$) so that the job-finding probability guarantees that the unemployment rate stays at 10.00%.

In conclusion, while the central bank successfully avoids the spiraling recession, the underlying composition of household savings inherently shifts. Employed and unemployed individuals simply adapt their distribution of assets along the new restricted grid space at the lower “natural” real interest rate, yielding a smooth transitional stationary equilibrium.

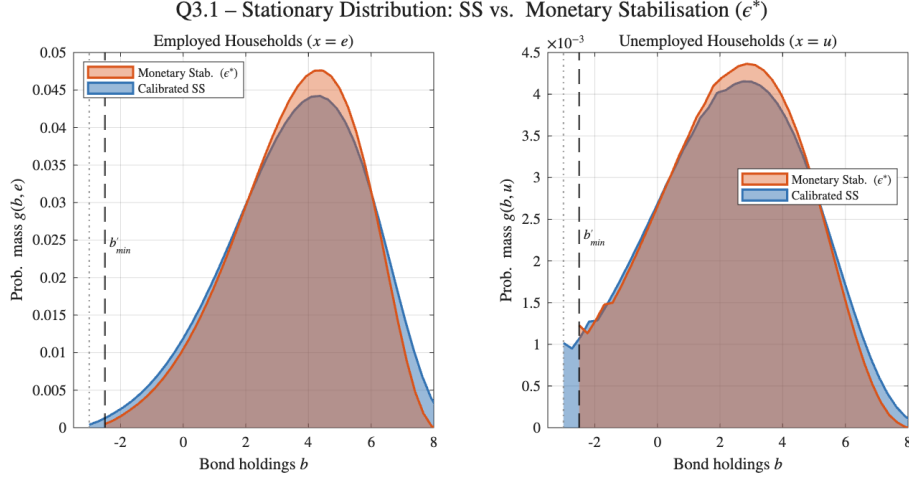


Figure 5: Asset Distribution: Steady State vs. Monetary Stabilization. Comparison of distributions before and after the shock under the optimal adjustment of the interest rate.

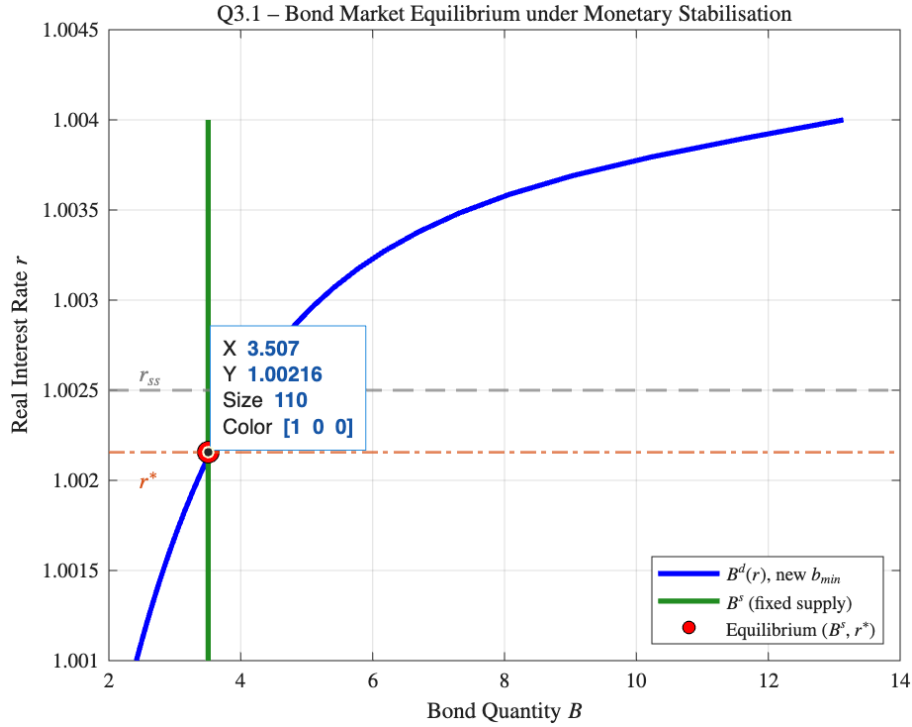


Figure 6: Bond Market: Demand $B^d(r)$ and Fixed Supply B^s . The figure illustrates the downward-sloping demand curve where the equilibrium natural rate falls from r_{ss} to r^* to clear the market.

Interpretation: The tightening of the borrowing constraint affects bond demand, which, ceteribus paribus, increases. Thus, the real interest rate must fall to the point that the total demand for bonds decreases until it matches the fixed bond supply again. Figure 6 maps total bond demand given the new borrowing constraint, the bond supply, and the real interest rate. However, although the labor market is insulated, the same cannot be said about the distribution of bonds. We observe in figure 5 that more workers bunch around bond holdings of 4. This is the result of the two counteracting effects: the households that were previously further ahead

in the distribution decrease their holdings due to the fall in the interest rate; for them, the effect of the fall in r is stronger than the tightening of the constraint. The same reasoning, but inverted, can be applied to the households that were instead closer to the bottom, who increase their holdings.

3.b) Offsetting the Financial Shock with Fiscal Policy

The Logic of the Optimal Fiscal Policy (B'_s)

Similar to the monetary policy response, the government can completely insulate the real economy from the permanent financial shock by perfectly stabilizing the labor market.

To prevent a recession, the labor market tightness θ must remain at its steady-state value of 1, which implies an unchanged unemployment rate ($u = 10\%$) and an unchanged firm surplus ($A - w$). This requires the real wage w to remain at its original steady-state level (w_{ss}).

Given the wage rigidity mechanism $w = w_{ss} \left(\frac{\pi_{ss}}{\pi} \right)^\xi$, inflation must be completely stabilized at its steady-state target: $\pi = \pi_{ss}$.

Because there is no monetary policy shock in this scenario ($\epsilon = 1$), the Taylor rule dictates the nominal interest rate:

$$i = i_{ss} \left(\frac{\pi}{\pi_{ss}} \right)^{\phi_\pi} \implies i = i_{ss}(1)^{\phi_\pi} = i_{ss} \quad (17)$$

If the nominal interest rate remains at i_{ss} and inflation remains at π_{ss} , the Fisher equation ($i = r\pi$) mechanically forces the real interest rate to remain exactly at its original steady-state value: $r = r_{ss}$.

However, the tighter borrowing limit ($b'_{min} = -2.5$) increases households' precautionary demand for savings ($B^d > B^s$). To clear the bond market without forcing the real interest rate to drop (which would trigger the recessionary loop), the government must exactly accommodate this higher demand for savings by issuing more public debt. The new bond supply B'_s must simply equal the new aggregate bond demand evaluated at the original steady-state interest rate r_{ss} .

Computational Strategy and Pseudo-Code

To computationally find the necessary B'_s , we design an algorithm (`main4_Bs.m`) that operates completely at the original steady-state prices:

1. **Setup:** We artificially “freeze” the labor market, real wages, and real interest rate at their initial steady-state values ($u = 10\%$, $w = w_{ss}$, $\theta = 1$, $r = r_{ss}$).
2. **Household Problem:** We set up the household EGM problem with the newly tightened borrowing limit ($b'_{min} = -2.5$) and solve for the optimal policy functions under the original steady-state prices.
3. **Stationary Distribution:** We simulate the new stationary distribution $g(b, x)$ using Young's (2010) method.

4. **Market Clearing:** We calculate the new aggregate bond demand B^d from this distribution, and simply set the required fiscal policy as $B'_s = B^d$. The difference $B'_s - B_s$ represents the issuance of new government bonds.

Results and Discussion

The EGM converges in 689 iterations. Under the new borrowing limit of -2.5 , precautionary savings behaviors increase bond demand significantly.

The new aggregate bond demand (B^d) evaluates to 3.994239. To completely accommodate this increased demand, the government sets $B'_s = 3.994239$.

Compared to the original steady-state bond supply of 3.507003, the government must issue 0.487236 additional bonds to soak up the increased precautionary savings demand. Because the government actively provides the exact amount of safe assets households desire at the current prices, the real interest rate r faces no downward pressure and remains exactly at $r_{ss} = 1.0025$.

Consequently, the entire nominal block is undisturbed. Inflation stays perfectly anchored at π_{ss} , the real wage w never surges, and firm profits remain stable. Firms maintain their vacancy posting ($\theta = 1$), and the unemployment rate stays perfectly at 10.00%. The fiscal intervention perfectly circumvents the recession without requiring any price or wage adjustments simply by expanding the supply of government debt to match the aggregate shift in household risk aversion.

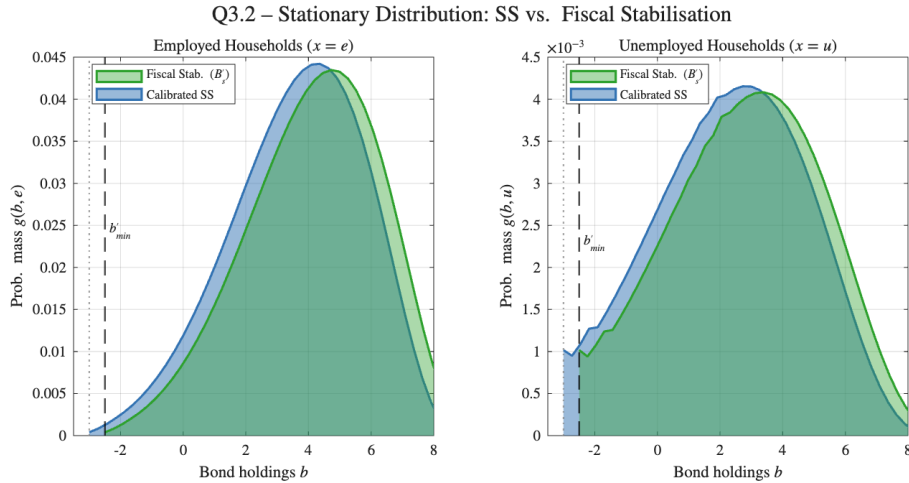


Figure 7: Asset Distribution: Steady State vs. Fiscal Stabilization. Comparison of distributions before and after the shock under the condition of expanded bond supply.

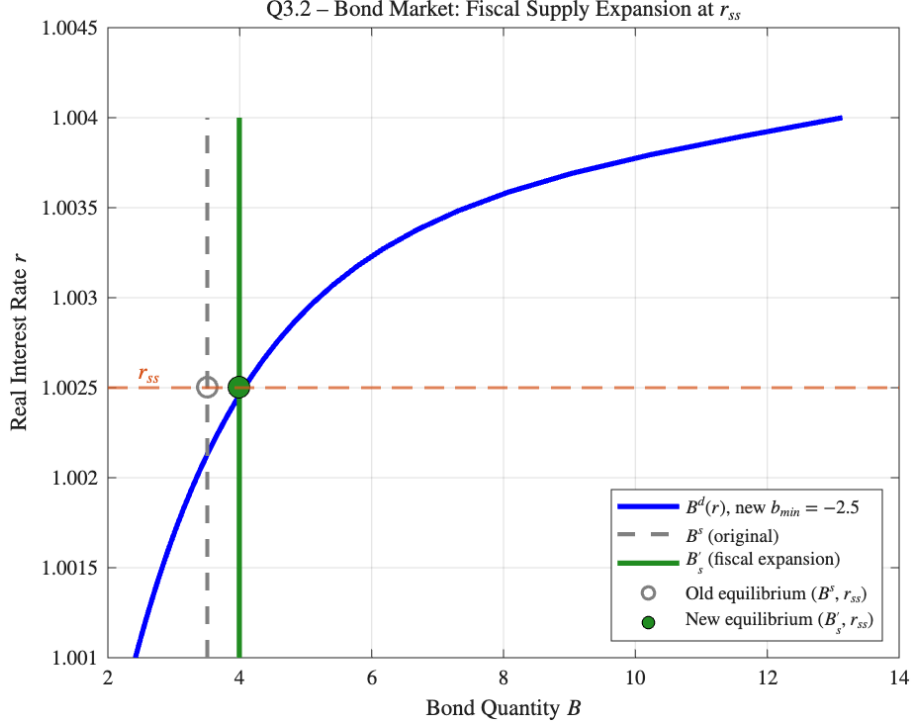


Figure 8: Bond Market: Demand $B^d(r)$ and the Fiscal Supply Shift. The government shifts the supply curve upward from B^s to B'_s to absorb the increased demand and maintain the interest rate at r_{ss} .

Interpretation: The effect the issuance of more bonds has on the distribution is quite straight-forward. As we discussed the government issues bonds until the market clears, keeping the real interest rate unchanged. The households are then able to satisfy their need for more bonds, and so they distribute them among each other, resulting in the bond holding distribution in figure 7 to approximately shift by the new bond issuance. Furthermore, figure 8 shows the new market-clearing equilibrium achieved through the fiscal expansion; the shift of the bond supply allows for a new equilibrium which keeps the interest rate at its steady state level.

3.c) The Zero-Lower-Bound and the Tightening of the Borrowing Constraint

The Zero Lower Bound constraint (b'_{min})

The final scenario explores the boundaries of monetary policy by finding the exact size of the financial shock (i.e. the limit borrowing constraint) that pushes the economy to the Zero Lower Bound (ZLB), where the nominal interest rate hits $i = 1$.

Under strict inflation targeting (we do not have a Taylor Rule anymore), the central bank stabilizes inflation to its target: $\pi = \pi_{ss}$. Consequently, if the nominal interest rate is forced down to the ZLB ($i = 1$), the Fisher equation ($i = r\pi$) strictly determines the resulting real interest rate:

$$1 = r\pi_{ss} \implies r_{zlb} = \frac{1}{\pi_{ss}} \quad (18)$$

Given our estimated inflation target ($\pi_{ss} = 1.001667$), this implies $r_{zlb} = 0.998336$. Because the

required real interest rate is strictly lower than the initial steady-state rate ($r_{ss} = 1.002500$), the aggregate demand for savings must rise significantly in order for the bond market to clear at this new rate.

We hypothesize that an extreme tightening of the borrowing constraint (a severe financial shock to some new limit b'_{min}) will force households to aggressively increase their precautionary savings, effectively driving the "natural" interest rate of the economy down to exactly r_{zlb} .

Computational Strategy

To computationally find the necessary b'_{min} , we design an algorithm (`main5.zlb.m`) that operates in the following loop, utilizing an iterative update to close the market clearing error:

1. **Setup:** We fix the real interest rate to the analytically derived $r_{zlb} = 1/\pi_{ss}$, and maintain the labor market variables at their initial steady-state values (as strict inflation targeting insulates the real wage and firm surplus).
2. **Initialize Guess:** We start with an initial guess for the financial shock (e.g., $b'_{min} = -2.5$).
3. **Household Problem:** For the current b'_{min} guess, we update the asset grid space and solve the EGM for the optimal policy functions at the fixed r_{zlb} .
4. **Stationary Distribution:** We simulate the new stationary distribution $g(b, x)$ using Young's (2010) method.
5. **Market Clearing:** We calculate the new aggregate bond demand B^d and compute the error $B^s - B^d$. If the demand is too low, we tighten the borrowing constraint further (pushing b'_{min} closer to zero) through an updating step $b'_{min}^{new} = b'_{min}^{old} + \lambda(B^s - B^d)$. We iterate until the bond market clears.

Results and Discussion

Executing this algorithm, the loop converges rapidly in 16 iterations. To push the nominal interest rate down precisely to $i = 1$ and clear the bond market at the required $r_{zlb} = 0.998336$, the borrowing constraint must tighten drastically to:

$$b'_{min} = -0.111872 \tag{19}$$

Discussion: Compared to the initial financial shock explored in Question 2 (which tightened the constraint from -3 to -2.5), dragging the economy all the way to the Zero Lower Bound requires a tight constraint of -0.1119 . At this nearly-zero boundary, the uninsurable idiosyncratic employment risk for the households becomes immense. To prevent large consumption drops upon unemployment when practically no borrowing is allowed, households vastly increase their precautionary buffer-stock savings.

This immense demand for safe assets is what ultimately forces the market-clearing real interest rate down to 0.998336 . Because the central bank is strictly maintaining $\pi = \pi_{ss}$, the nominal

interest rate is mathematically pulled down in tandem until it bottoms out at exactly $i = 1$. This finding highlights the severe size of the financial friction necessary to exhaust conventional monetary policy room in this HANK-SAM framework.

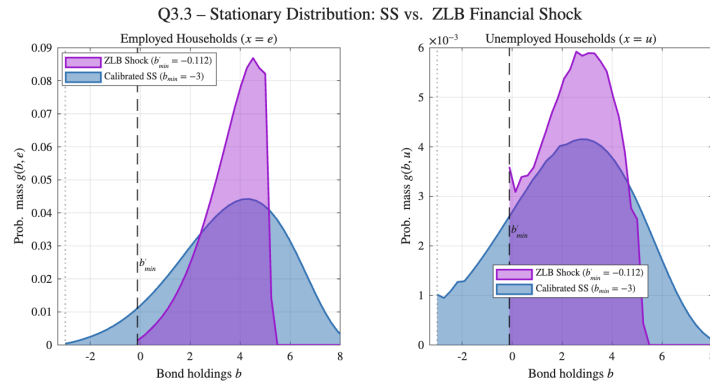


Figure 9: Asset Distribution: Steady State vs. ZLB Shock. Stationary distributions before and after the extreme shock to $b'_{min} = -0.1119$.

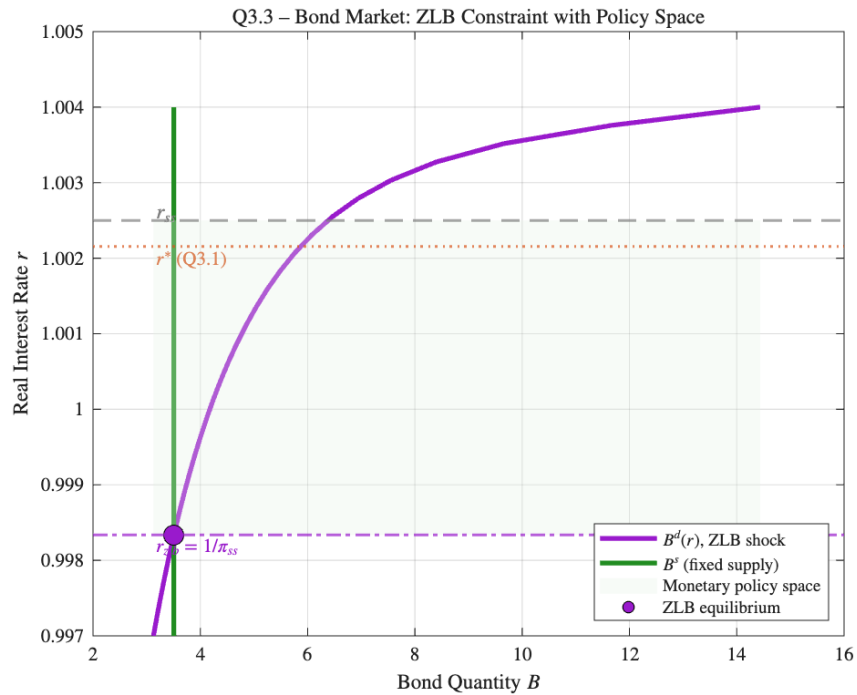


Figure 10: Bond Market at the ZLB: Demand $B^d(r)$ with annotated policy space. The figure shows the demand curve, fixed supply B^s , and vertical lines at r_{ss} , r^* , and r_{zlb} .

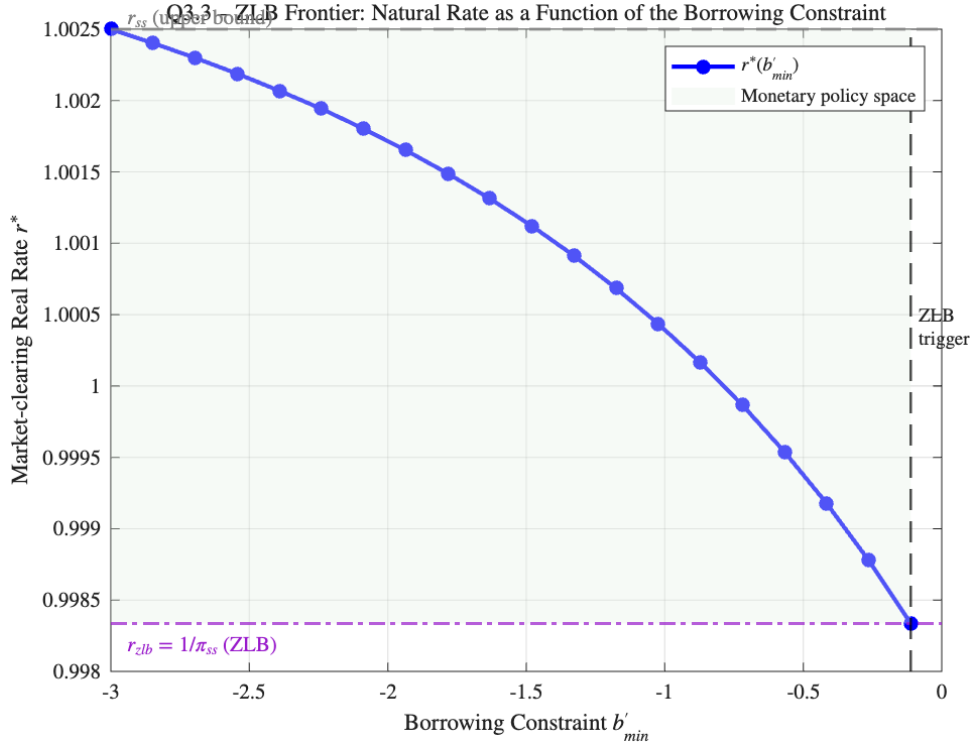


Figure 11: Q3.3 – ZLB Frontier: natural real rate $r^*(b'_{min})$ as a function of the borrowing constraint. The curve traces the market-clearing real interest rate for each possible financial shock magnitude, from the initial $b_{min} = -3$ (right) to the ZLB-triggering $b'_{min} = -0.1119$ (left). Dashed horizontal lines mark r_{ss} (upper bound of monetary policy space) and $r_{zlb} = 1/\pi_{ss}$ (lower bound). The vertical dashed line marks the exact shock required to exhaust conventional monetary policy.

Interpretation: Figures 9, 10, 11 provide a visual overview of what happens to the economy once we hit the ZLB. For both workers and unemployed, the effect is quite dramatic: with such a tight borrowing constraint, the precautionary motive becomes huge. Households will want to buy more bonds to self insure. However, many unemployed households will not be able to afford more bonds and will be forced at the borrowing constraint. Furthermore, because the gross real interest rate has fallen below 1 ($r_{zlb} < 1$), the return on savings effectively acts as a wealth tax. Even though households have an extreme precautionary desire to save due to the strict borrowing limit, the negative net return mathematically prevents them from accumulating large buffer stocks. Figure 9 shows that virtually no household holds more than 6 bonds. Figure 10 shows the market-clearing equilibrium at the ZLB. This shows us how the tightening makes the households "desperate": their demand clears the market even when they are effectively losing money by keeping the bonds. Concluding, figure 11 maps the market-clearing interest rate as a function of the borrowing constraint. It is easy to see that the tighter the constraint is, the lower the interest rate required to clear the bond market.

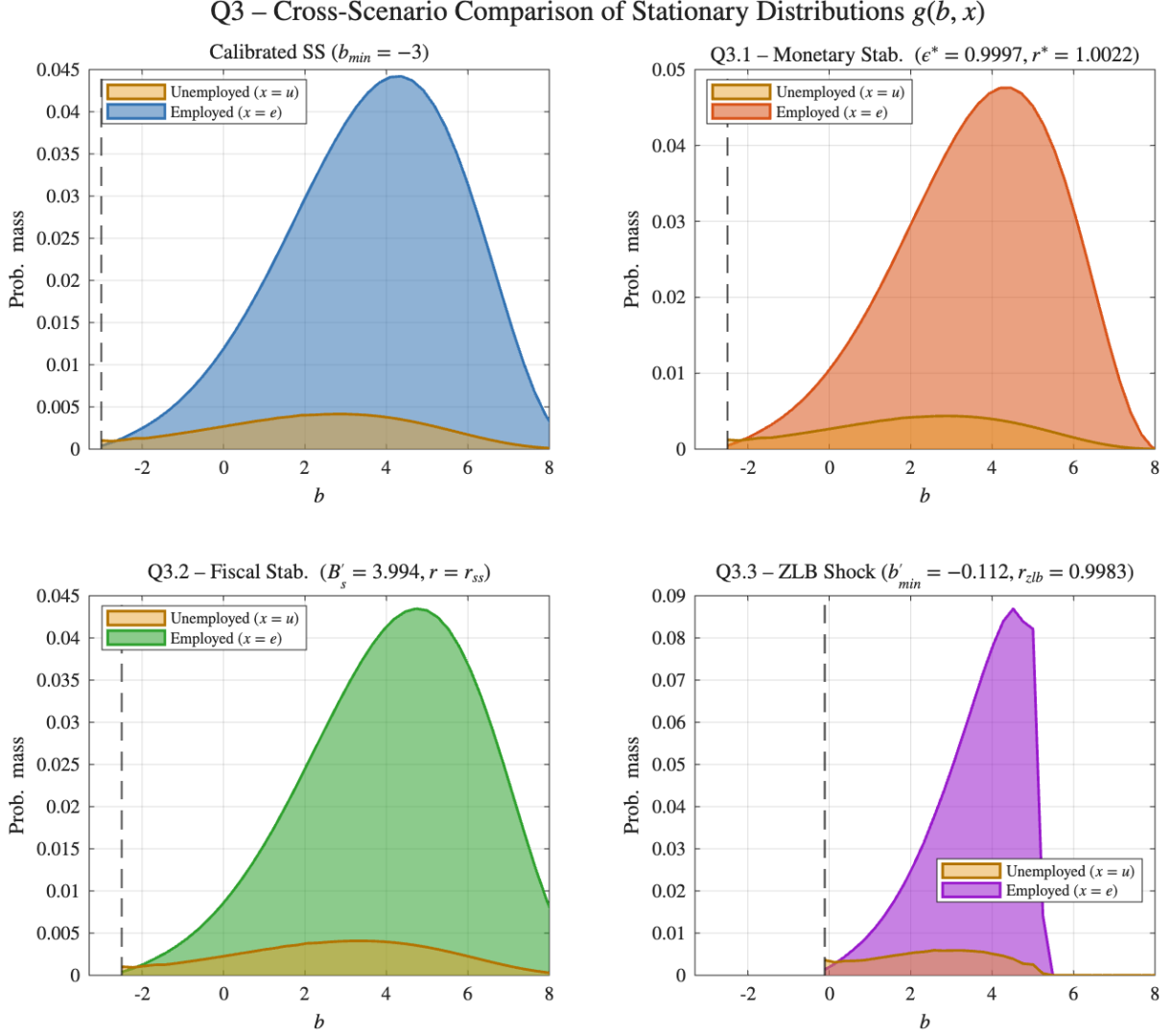


Figure 12: Cross-scenario comparison of stationary distributions $g(b, x)$. All panels use the same axes for direct comparison. Solid lines: employed households; dashed lines: unemployed households. *Top-left*: original calibrated steady state. *Top-right and bottom-left*: both policy responses stabilize the distribution relative to the unchecked shock. *Bottom-right*: the extreme ZLB shock drastically compresses the left tail of the distribution.

Interpretation: The panel shows the distributions across the four cases. We observe that, excluding the ZLB case, the distribution is almost unchanged. Although the distribution does change as a function of the policy implemented, the visual outcome confirms that the policies successfully counterbalance the tightening of the borrowing constraint.

Moreover, we can note the similarity between Q3.1 and Q3.3: the ZLB case is effectively the extremization of the case of financial shock. Because of this, the figure of Q3.3 shows the shape the distribution would have tended to had the tightening of the borrowing constraint been even stronger.

Conclusion: The Failure of Naive Taylor Rules

The results from our monetary stabilization exercise and our ZLB analysis illustrate a fundamental weakness of relying solely on simple Taylor rules in a macroeconomy characterized by heterogeneous agents and search frictions. In short, naive Taylor rules fail because they do not track the natural real interest rate and ignore the demand-amplifying effects of endogenous countercyclical income risk.

1. The Flaw in the Simple Taylor Rule: Under a standard Taylor rule, the central bank only lowers the nominal interest rate i in response to falling inflation π . This creates a dangerous delay: the central bank waits for aggregate demand to collapse before providing accommodation. In a HANK-SAM model, this initial drop in demand reduces firm vacancies and lowers the job finding rate. A lower job finding rate increases unemployment risk, which triggers a spike in precautionary saving. This downward spiral means the real interest rate needs to fall even further to stabilize the economy. If the Taylor rule is too passive (e.g., $\phi_\pi = 1.2$), the economy falls into a catastrophic recession with unemployment reaching over 80%.

2. The Necessity of Active Monetary Policy: Our finding from the monetary policy response shows that to perfectly insulate the labor market, the central bank must shift the Taylor rule intercept using an exogenous monetary shock ($\epsilon = 0.999657$). By adjusting ϵ to match the natural rate r^* , the central bank accommodates the households' increased precautionary savings demand without altering inflation. This prevents the initial drop in aggregate demand, short-circuiting the endogenous risk feedback loop before firm surplus is destroyed. A naive Taylor rule cannot do this because its intercept is rigidly fixed.

3. The Extreme Limits of the ZLB: Furthermore, our ZLB analysis proves how difficult stabilization is when prices are perfectly sticky or the central bank strictly targets inflation ($\pi = \pi_{ss}$). To actually drive the nominal interest rate down to $i = 1$ under this regime, it requires a large tightening of the borrowing constraint to $b'_{min} = -0.111872$. This highlights that when the central bank stabilizes prices, the real interest rate carries the entire burden of clearing the asset market. If the central bank were instead constrained by a naive Taylor rule, the required deflation to hit the ZLB would heavily increase the real wage (due to nominal wage rigidities), obliterating firm surplus and sending unemployment rising significantly.